

## 1 · POST-MONEY vs PRE-MONEY VALUATION (VC Method)

**Real Talk:** Pre-money = what investors think the company is worth BEFORE they put money in. Post-money = pre-money + new investment. Investor ownership % = investment ÷ post-money. Founders own pre-money ÷ post-money. Students always trip up on which denominator to use.

**Investor % = Investment ÷ Post-Money | Founder % = Pre-Money ÷ Post-Money | Post-Money = Pre-Money + Investment**

### WORKED EXAMPLE — VC METHOD

**Given:** Pre-money = \$4M | Investment = \$1M | 8M founder shares outstanding

#### Q1 – Founder % and Investor % after this round?

Step 1: Post-Money = \$4M + \$1M = **\$5M**

*Intuition: Investor is buying a slice of the pie – add their money to the pie's total value.*

Step 2: Investor % = \$1M ÷ \$5M = **20%**

Step 3: Founder % = \$4M ÷ \$5M = **80%**

#### Q2 – Price per share (PPS)? Founder shares = 8M.

Step 1: Total shares = Founder shares ÷ Founder % = 8M ÷ 80% = **10M shares**

*Intuition: If founders own 80% and have 8M shares, 8M must be 80% of total.*

Step 2: PPS = Post-Money ÷ Total shares = \$5M ÷ 10M =

**\$0.50/share**

## 2 · SAFE NOTES — Valuation Cap vs Discount

**Real Talk:** A SAFE is NOT a loan — no interest, no maturity date. It gives investors the right to buy shares at a 优惠 price when the Series A closes. The cap sets a CEILING on the price per share. The discount sets a percentage OFF the Series A price. The investor gets whichever price is LOWER (more shares for their money).

**SAFE Price = MIN(Cap ÷ Pre-SAFE Shares, Series A Price × (1 – Discount%))**

### WORKED EXAMPLE — SAFE CAP VS DISCOUNT

**Given:** SAFE cap = \$5M | Discount = 20% | Series A pre-money = \$8M | 10M pre-SAFE shares

#### Step 1 – Calculate the cap price per share:

Cap price = \$5M ÷ 10M = **\$0.50/share**

*Intuition: The SAFE investor will never pay more than \$0.50/share regardless of Series A pricing.*

#### Step 2 – Calculate the discounted Series A price:

Series A price = \$8M ÷ 10M = \$0.80/share

Discounted price = \$0.80 × (1 – 0.20) = \$0.80 × 0.80 =

**\$0.64/share**

*Intuition: Discount gives 20% off the Series A price – but cap might be lower.*

#### Step 3 – Choose the LOWER price (more shares for investor):

SAFE Price = MIN(\$0.50, \$0.64) = **\$0.50/share** ← Cap wins

vs. Series A investor who pays \$0.80/share

→ SAFE investor gets **37.5% more shares** for the same money

#### Step 4 – Who wins? Calculate ownership after conversion:

## 3 · CONVERTIBLE NOTES — Interest + Cap + Discount

**Real Talk:** Unlike SAFE, convertible notes ARE loans — they accrue interest and have a maturity date. At conversion, accrued interest is added to principal first, THEN the total is converted at the better of cap price or discounted price. If not converted by maturity, notes must be repaid as debt.

**Accrued Value = Principal × (1 + Rate%)^years | Conversion Price = MIN(Cap Price, Discount Price)**

#### WORKED EXAMPLE — CONVERTIBLE NOTE

**Given:** \$500K note | 8% p.a. | 2 years | Cap = \$4M | Discount = 25% | Series A pre-money = \$6M | 6M pre-note shares

##### Step 1 – Grow the principal with compound interest:

$$\text{Accrued value} = \$500K \times (1.08)^2 = \$500K \times 1.1664 =$$

**\$583,200**

*Intuition: Interest compounds on interest – Year 1: \$500K × 1.08 = \$540K. Year 2: \$540K × 1.08 = \$583,200.*

##### Step 2 – Calculate both conversion prices:

$$\text{Cap price} = \$4M \div 6M = \text{\textbf{\$0.667/share}}$$

$$\text{Discounted Series A} = (\$6M \div 6M) \times (1 - 0.25) = \$1.00 \times 0.75 = \text{\textbf{\$0.75/share}}$$

*Intuition: Cap price = ceiling on valuation per share.  
Discount price = Series A price minus discount.*

##### Step 3 – Choose the LOWER price (more shares):

$$\text{Conversion price} = \text{MIN}(\$0.667, \$0.75) = \text{\textbf{\$0.667/share}} \leftarrow \text{Cap wins}$$

##### Step 4 – How many shares does the note convert to?

$$\text{Shares} = \$583,200 \div \$0.667 = \text{\textbf{874,000 shares}}$$

*Intuition: More accrued interest → more shares at conversion.*

#### 4 · DILUTION — Option Pool & New Round

**Real Talk:** Dilution = new shares issued reduce everyone's ownership %. The option pool (ESOP) is typically created BEFORE a new investor enters — it comes out of founder shares. Both founders AND the option pool dilute pro-rata when new money arrives. The option pool does NOT shield founders — it's just another holder.

$$\text{Investor \%} = \text{New Shares} \div (\text{Existing} + \text{New}) \quad | \quad \text{Post-Dilution Founder \%} = \text{Pre} \times (1 - \text{Dilution\%})$$

#### WORKED EXAMPLE — DILUTION WITH OPTION POOL

**Given:** Pre-round: Founders 8M shs (80%) | ESOP 2M shs (20%) | Total 10M | New investor: \$3M at \$1.50/share → 2M new shares

##### Pre-round ownership – confirm the baseline:

$$\text{Founders: } 8M \div 10M = \text{\textbf{80\%}} \quad | \quad \text{ESOP: } 2M \div 10M = \text{\textbf{20\%}}$$

*Intuition: The ESOP is already issued – it's a real holder like any other shareholder.*

##### Step 1 – Calculate new total shares after investment:

$$\text{New total} = 10M + 2M = \text{\textbf{12M shares}}$$

*Intuition: Adding new shares always increases total – existing holders get diluted.*

##### Step 2 – Investor ownership %:

$$\text{Investor \%} = 2M \div 12M = \text{\textbf{16.67\%}}$$

##### Step 3 – Dilution rate applied to each existing holder:

$$\text{Total dilution rate} = 2M \div 12M = \text{\textbf{16.67\%}}$$

$$\text{Founder dilution} = 80\% \times (1 - 0.1667) = \text{\textbf{66.67\%}}$$

$$\text{ESOP dilution} = 20\% \times (1 - 0.1667) = \text{\textbf{16.67\%}}$$

$$\text{New investor} = \text{\textbf{16.67\%}}$$

#### 5 · CAP TABLE — Full 4-Round Walkthrough

**Real Talk:** A cap table tracks ownership % as a company raises across multiple rounds. Pre-money is the company's value before the new round. Price per share = pre-money ÷ pre-round total shares. New shares = investment ÷ PPS. Work round by round — always track cumulative shares.

#### WORKED EXAMPLE — FULL CAP TABLE

**Given:** Founders: 8M shs @ \$0.05 | Seed: \$500K at \$0.20/sh | Series A: \$2M at \$0.60/sh | Series B: \$4M at \$1.60/sh

##### Round 0 – Pre-Seed (Founders own 100%):

$$\text{Price per share (PPS)} = \$0.05 \quad | \quad \text{Total shares} = \text{\textbf{8M}}$$

$$\text{Company value} = 8M \times \$0.05 = \$400K$$

##### After Seed (\$500K at \$0.20/share):

$$\text{New shares} = \$500K \div \$0.20 = \text{\textbf{2.5M}}$$

$$\text{Total} = 8M + 2.5M = \text{\textbf{10.5M}}$$

$$\text{Founders: } 8M \div 10.5M = \text{\textbf{76.2\%}} \quad | \quad \text{Seed: } 2.5M \div 10.5M = \text{\textbf{23.8\%}}$$

##### After Series A (\$2M at \$0.60/share):

$$\text{New shares} = \$2M \div \$0.60 = \text{\textbf{3.333M}}$$

$$\text{Total} = 10.5M + 3.333M = \text{\textbf{13.833M}}$$

$$\text{Founders: } \text{\textbf{57.8\%}} \quad | \quad \text{Seed: } \text{\textbf{18.1\%}} \quad | \quad \text{Series A: } \text{\textbf{24.1\%}}$$

##### After Series B (\$4M at \$1.60/share):

$$\text{New shares} = \$4M \div \$1.60 = \text{\textbf{2.5M}}$$

$$\text{Total} = 13.833M + 2.5M = \text{\textbf{16.333M}}$$

$$\text{Founders: } \text{\textbf{49.0\%}} \quad | \quad \text{Seed: } \text{\textbf{15.3\%}} \quad | \quad \text{Series A: } \text{\textbf{20.4\%}} \quad |$$

#### 6 · LIQUIDATION PREFERENCE — 1× vs 2× Participating

**Real Talk:** Liquidation preference = who gets paid first when the company is sold. Non-participating investors choose the better of their preference OR pro-rata share. Participating investors get BOTH — preference first, then their share of the remainder pro-rata. Participating is much more valuable for investors at large exits.

$$\text{1× Non-Part: } \text{Max}(\text{Preference, Pro-Rata}) \quad | \quad \text{Participating: } \text{Preference} + (\text{Remaining} \times \text{Ownership\%})$$

#### WORKED EXAMPLE — LIQUIDATION PREFERENCE



## 7 · IRR — Internal Rate of Return

**Real Talk:** IRR is the annual return rate that makes NPV of all cash flows = zero. Think of it as: "What interest rate would I need at the bank to get the same result?" Higher IRR = better return. VCs target 25–35% IRR. For single investment + exit with no interim flows:  $IRR = (MOIC)^{1/Years} - 1$ .

$$\text{Simple IRR} = (\text{Exit} \div \text{Investment})^{1 \div \text{Years}} - 1 \quad | \quad NPV = \sum CF_t \div (1+IRR)^t = 0$$

### WORKED EXAMPLE — IRR (SIMPLE & WITH INTERIM CASH FLOWS)

**Given:** Invest \$2M | Exit \$10M | 5-year hold | Interim: Yr2 +\$1M, Yr5 +\$5M

#### Q1 – Simple IRR (no interim cash flows):

Step 1:  $MOIC = \$10M \div \$2M = 5.0\times$

*Intuition:* 5× return on investment over 5 years.

Step 2:  $IRR = (5.0)^{1 \div 5} - 1 = 5.0^{0.20} - 1$

Step 3:  $5^{0.20} = 1.3797 \rightarrow IRR = 37.97\% \approx 38\%$

#### Q2 – With interim cash flows: Yr0: -\$2M | Yr2: +\$1M | Yr5: +\$5M

Step 1: Try IRR = 20%:

$$NPV = -2 + 1/(1.20)^2 + 5/(1.20)^5$$

$$= -2 + 0.694 + 2.01 = +0.70 \text{ (NPV > 0, IRR > 20\%)}$$

Step 2: Try IRR = 30%:

$$NPV = -2 + 1/(1.30)^2 + 5/(1.30)^5$$

$$= -2 + 0.591 + 1.35 = -0.06 \text{ (NPV < 0, IRR < 30\%)}$$

Step 3: IRR is between 20% and 30%, closer to 20% → ≈ 28%

28%

## 8 · DCF — Discounted Cash Flow with Terminal Value

**Real Talk:** DCF values a company by predicting future cash flows and discounting them back to today. Money today is worth more than money tomorrow (opportunity cost). The discount rate = WACC. Terminal value captures everything after the forecast period — usually 70–85% of total DCF value. Get WACC right and get g (growth rate) right.

$$\text{Enterprise Value} = \sum PV(FCF_t) + PV(\text{Terminal Value}) \quad | \quad TV = FCF_n \times (1+g) \div (WACC-g)$$

### WORKED EXAMPLE — DCF (3-YEAR FORECAST)

**Given:** FCF Y1=\$100K, Y2=\$120K, Y3=\$150K | WACC=12% | g=3% | Terminal FCF = Y3 × (1+g)

#### Step 1 – Calculate Terminal Value (at end of Year 3):

$$TV = FCF_3 \times (1+g) \div (WACC - g)$$

$$TV = \$150K \times 1.03 \div (0.12 - 0.03) = \$154.5K \div 0.09 =$$

$$\boxed{\$1,717K}$$

*Intuition:* After Year 3 we assume FCF grows forever at g%. The Gordon Growth Model capitalizes that perpetual stream.

#### Step 2 – Discount each FCF to present value:

$$PV(Y1) = \$100K \div 1.12^1 = \$100K \div 1.12 = \boxed{\$89.3K}$$

$$PV(Y2) = \$120K \div 1.12^2 = \$120K \div 1.2544 = \boxed{\$95.7K}$$

$$PV(Y3) = \$150K \div 1.12^3 = \$150K \div 1.4049 = \boxed{\$106.8K}$$

*Intuition:* Discounting "undoes" compounding — later cash flows are worth less today.

#### Step 3 – Discount Terminal Value back to today:

$$PV(TV) = \$1,717K \div 1.12^3 = \$1,717K \div 1.4049 = \boxed{\$1,222.2K}$$

*Intuition:* TV is the biggest component. It's discounted by 3 years of WACC.

## 9 · WACC — Weighted Average Cost of Capital

**Real Talk:** WACC is the average interest rate a company pays on ALL its capital — both equity and debt. It's the discount rate in DCF. Equity holders demand higher returns than debt holders (more risk), so equity is more expensive. The after-tax cost of debt is lower because interest is tax-deductible.

$$WACC = (E \div V \times Re) + (D \div V \times Rd \times (1-T)) \quad | \quad E=Equity, D=Debt, V=E+D, T=Tax \text{ rate}$$

### WORKED EXAMPLE — WACC

**Given:** Equity = \$8M | Debt = \$2M | Re = 18% | Rd = 8% | Tax = 25%

#### Step 1 – Find capital weights:

$$\text{Total capital } V = \$8M + \$2M = \$10M$$

$$E/V = \$8M \div \$10M = 80\% \text{ (equity weight)}$$

$$D/V = \$2M \div \$10M = 20\% \text{ (debt weight)}$$

*Intuition: Weight by market value, not book value. Debt % matters because interest is deductible.*

#### Step 2 – Calculate each cost component:

$$\text{Equity cost} = 80\% \times 18\% = 14.4\%$$

$$\text{After-tax debt cost} = 20\% \times 8\% \times (1 - 0.25)$$

$$= 20\% \times 8\% \times 0.75 = 20\% \times 6\% = 1.2\%$$

*Intuition: Tax shield: company saves 25% of interest cost. Without tax:  $20\% \times 8\% = 1.6\%$ . With tax: 1.2%.*

#### Step 3 – Sum components:

$$WACC = 14.4\% + 1.2\% = 15.6\%$$

**Exam tip:** If the company has no debt, WACC = Re (just the cost of equity). If the tax

## 10 · EVA — Economic Value Added (NOPAT Approach)

**Real Talk:** EVA answers: "Is the company actually creating wealth for shareholders?" A company can be profitable on paper but still destroy value if it doesn't earn more than the cost of ALL capital (debt + equity). EVA = NOPAT – Capital Charge. Positive EVA = management created value. Negative EVA = they destroyed value despite accounting profits.

$$EVA = NOPAT - (\text{Capital} \times WACC) \quad | \quad NOPAT = \text{Operating Profit} \times (1 - \text{Tax Rate})$$

### WORKED EXAMPLE — EVA (NOPAT APPROACH)

**Given:** Operating profit = \$1,200 | Capital (debt + equity) = \$9,000 | WACC = 10% | Tax rate = 40%

#### Step 1 – Convert accounting profit to NOPAT (true economic profit):

$$NOPAT = \text{Operating Profit} \times (1 - \text{Tax Rate})$$

$$NOPAT = \$1,200 \times (1 - 0.40) = \$720K$$

*Intuition: NOPAT = what operating profit would be if there were no interest or tax effects – pure business profitability before financing.*

#### Step 2 – Calculate the capital charge (opportunity cost of all capital):

$$\text{Capital charge} = \text{Capital} \times WACC = \$9,000 \times 10\% = \$900K$$

*Intuition: \$9M of capital costs \$900K/year to finance – this is what investors demand.*

#### Step 3 – Subtract capital charge from NOPAT:

$$EVA = \$720K - \$900K = (-\$180K)$$

## 11 · CONTRIBUTION MARGIN & SUSTAINABLE GROWTH RATE

**Real Talk:** Contribution Margin = how much each dollar of sales contributes to covering fixed costs after variable costs are paid. It's the "leftover" after the variable cost of making the product. Sustainable Growth Rate (SGR) = max growth rate using only internal funds (retained earnings) — no new equity or debt.

$$CM\% = (\text{Sales} - \text{Variable Costs}) \div \text{Sales} \quad | \quad SGR = ROE \times \text{Retention Ratio} \quad | \quad \text{Break-even} = \text{Fixed Costs} \div CM\%$$

### WORKED EXAMPLE — CONTRIBUTION MARGIN + SGR + BREAK-EVEN

**Given:** Cash fixed costs = \$60K | Variable costs = \$70K | Sales = \$100K | Net Income = \$20K | ROE = 15%

#### Q1 – Contribution Profit Margin %:

$$\text{Step 1: Contribution} = \text{Sales} - \text{Variable Costs} = \$100K - \$70K = \$30K$$

$$\text{Step 2: CM\%} = \$30K \div \$100K = 30\%$$

*Intuition: For every \$1 of sales, \$0.30 is available to cover fixed costs and profit.*

#### Q2 – Break-even point in sales dollars:

$$\text{Break-even} = \text{Fixed Costs} \div CM\% = \$60K \div 0.30 = \$200K \text{ in sales}$$

*Intuition: Need \$200K in sales to exactly cover all costs (zero profit, zero loss).*

#### Q3 – Sustainable Growth Rate:

## 12 · TERMINAL VALUE & COMPARABLE COMPANY VALUATION

**Real Talk:** Terminal Value (TV) = value of all cash flows beyond the forecast period, captured in a single number today. Gordon Growth Model assumes cash flows grow forever at rate g. Comparable company valuation uses P/E multiples — find a similar public company's P/E ratio and apply it to the target's EPS.

$$TV = FCF_n \times (1+g) \div (WACC-g) \quad | \quad \text{Target Value} = \text{Target EPS} \times (\text{Comp Price} \div \text{Comp EPS})$$

### WORKED EXAMPLE — TERMINAL VALUE + COMPARABLE COMPANY

**Given:** Current net income = \$20K | Next year FCF = \$26K | g = 7% | Required return = 20% | Comp: Price=\$20, EPS=\$2 (not \$3), Net Income=\$20K, Shares=10K

#### Q1 – Terminal Value (Gordon Growth Model):

Step 1: Confirm Year 1 FCF = \$26K (next year's cash flow



## 13 · TERM SHEET CLAUSES — What Each Really Means

**Real Talk:** A term sheet is the blueprint for a VC deal before lawyers draft legal documents. Each clause has direct financial consequences for founders and investors. Anti-dilution protects investors from bad future valuations. Vesting keeps founders working. Board control determines who runs the company.

|                                |                                                                                                                                                                                                                                                                              |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Anti-Dilution (Ratchet)</b> | Broad-based weighted avg: uses ALL shares (including options) to calculate new price. Full-ratchet: replaces ALL shares at the new low price. Both hurt founders when company raises at a lower valuation – they issue more shares to investors to "make up" the difference. |
| <b>Liquidation Pref (1x)</b>   | Investor gets 1x their money back FIRST. Non-participating = choose pref OR pro-rata (whichever is better). Participating = BOTH preference AND pro-rata share of remaining. Participating is most favorable to investors at large exits.                                    |
| <b>Pay-to-Play</b>             | Investors who skip future rounds lose anti-dilution protection or board seats. Prevents "free riders" who don't participate in new rounds. Keeps cap table clean and signals investor commitment.                                                                            |
| <b>Pro-Rata Rights</b>         | Investors can buy new shares in future rounds to maintain their %. Founders dislike this – dilutes their ability to lead rounds and signals investor distrust of founders' ability to fundraise.                                                                             |

## 14 · EXIT MATH — M&A vs IPO vs Secondary

**Real Talk:** How founders and investors actually get their money out. M&A = trade sale to another company. IPO = going public (180-day lock-up typically). Secondary = selling existing shares before a liquidity event. Earlier liquidity = lower required return because risk is reduced.

$$\text{IRR} = (\text{Exit Proceeds} \div \text{Investment})^{(1 \div \text{Years})} - 1 \quad |$$

$$\text{MOIC} = \text{Exit} \div \text{Investment}$$

### WORKED EXAMPLE — EXIT SCENARIOS

**Given:** Invest \$3M at \$10M pre-money | 7M shares outstanding | 3-year hold | Compare M&A, IPO, Secondary

#### Step 1 – Establish baseline: price per share at investment:

$$\text{PPS} = \text{Pre-money} \div \text{Shares} = \$10\text{M} \div 7\text{M} = \text{\textcolor{teal}{\$1.43/share}}$$

Investment buys:  $\$3\text{M} \div \$1.43 = \text{\textcolor{teal}{2.1M shares}}$  (30% of company)

#### Scenario A – M&A at \$20M (2x exit):

$$\text{Exit per share} = \$20\text{M} \div 7\text{M} = \$2.86/\text{share}$$

$$\text{MOIC} = \$20\text{M} \div \$10\text{M} = \text{\textcolor{teal}{2.0x}}$$

$$\text{IRR} = (2.0)^{(1 \div 3)} - 1 = 1.26 - 1 = \text{\textcolor{teal}{25.99\%} \approx 26\%}$$

*Intuition: 2x in 3 years = 26% IRR. Mediocre for VC but possible in a strategic sale.*

#### Scenario B – IPO at \$50M (5x exit):

$$\text{MOIC} = \$50\text{M} \div \$10\text{M} = \text{\textcolor{teal}{5.0x}}$$

$$\text{IRR} = (5.0)^{(1 \div 3)} - 1 = 1.71 - 1 = \text{\textcolor{teal}{70.6\%}}$$

*Intuition: 5x in 3 years = 71% IRR. This is an excellent outcome for early investors.*

#### Scenario C – Secondary sale at \$15M after Year 2:

Sell 30% of shares at \$15M pre-money → \$4.5M cash at Year

2

## 15 · VESTING SCHEDULES — The Cliff Explained

**Real Talk:** Vesting protects both sides. Investors don't want to pay founders who quit early. Founders want job security. The cliff = waiting period before ANY

shares vest. Nothing vests until month X, then a chunk vests all at once, then monthly after. Think of it as a staircase: flat → big step → gradual slope.

**4-year vesting standard | Cliff at 12 or 14 months | Monthly or quarterly vest thereafter**

#### WORKED EXAMPLE — VESTING AT MONTH 25 VS MONTH 6

**Given:** 4-year vesting (48 months) | 14-month cliff | 20% vested at cliff | Monthly thereafter | Compare: founder leaves at month 25 vs month 6

##### Step 1 – Map out the vesting schedule:

Total period = 48 months | Cliff at month 14 | 20% vests at cliff

Remaining 80% vests over months 15–48 = **34 months**

Monthly vest rate after cliff =  $80\% \div 34 \text{ months} = \mathbf{2.353\% \text{ per month}}$

*Intuition: After the cliff jump, you earn ~2.35% of your shares every month.*

##### Scenario A – Founder leaves at month 25:

Before cliff (months 1–14): **0%** vested – still in cliff period

At cliff (month 14): **20%** vests all at once

After cliff (months 15–25): 11 months  $\times$  2.353% = **25.88%**

Total vested = 20% + 25.88% = **45.88%**

*Intuition: At month 25, the founder has climbed 45.88% of the vesting staircase.*

##### Scenario B – Founder leaves at month 6 (before cliff):

## 16 · WORKING CAPITAL & LIQUIDITY RATIOS

**Real Talk:** Liquidity ratios answer: "Can we pay our short-term bills?" Current Ratio > 1.5–2 is healthy. Quick Ratio < 1 means even liquidating all receivables won't cover current debts — a warning sign. The Cash Conversion Cycle (CCC) measures how many days cash is tied up in the business cycle.

**NWC = CA - CL | Current Ratio = CA ÷ CL | Quick = (CA - Inventory) ÷ CL | CCC = DSO + DIO - DPO**

#### WORKED EXAMPLE — WORKING CAPITAL & CCC

**Given:** Cash=\$50K | A/R=\$80K | Inventory=\$40K | A/P=\$60K | Accrued Exp=\$30K | Current Tax=\$50K | Sales=\$500K | COGS=\$300K

##### Step 1 – Identify Current Assets and Current Liabilities:

CA = Cash + A/R + Inventory = \$50K + \$80K + \$40K = **\$170K**

CL = A/P + Accrued Exp + Current Tax = \$60K + \$30K + \$50K = **\$140K**

NWC = \$170K - \$140K = **\$30K** (positive = liquidity buffer)

*Intuition: NWC = net short-term liquidity. Positive = more short-term assets than liabilities.*

##### Step 2 – Liquidity Ratios:

Current Ratio = \$170K ÷ \$140K = **1.21x** (below 1.5 = slightly weak)

Quick Ratio =  $(\$170K - \$40K) \div \$140K = \$130K \div \$140K = \mathbf{0.93x}$

*Intuition: Quick ratio < 1.0 – even selling all receivables won't fully cover current debts. Alert!*

##### Step 3 – Cash Conversion Cycle:

DSO =  $(A/R \div Sales) \times 365 = (\$80K \div \$500K) \times 365 = \mathbf{58.4 \text{ days}}$

*Intuition: How many days to collect from customers after a sale?*

## 17 · ARR, CHURN, CAC/LTV, LTV:CAC & EPS

**Real Talk:** These are the metrics SaaS and subscription businesses live by. ARR = annual run rate of recurring revenue. Churn = revenue lost to cancellations. CAC = cost to acquire a new customer. LTV = lifetime profit per customer. LTV:CAC < 3 = burning money; > 5 = under-investing in growth.

**ARR = MRR  $\times$  12 | Net Rev Churn = (MRR Lost - MRR Expansion) ÷ Start MRR | CAC = S&M Spend ÷ New Customers**

**LTV = (ARPU  $\times$  Gross Margin%) ÷ Churn Rate | Diluted EPS = Net Income ÷ (Shares + Net Dilution from Options)**

#### WORKED EXAMPLE — ARR + CHURN + CAC/LTV + EPS

**Given:** Jan MRR=\$50K | Feb: +\$8K new, -\$3K churn | Sales=\$120K/yr | Mktg=\$80K/yr | 100 new customers | ARPU=\$500/mo | GM=70% | Churn=3%/mo | Net Income=\$1M | Shares=500K | Options=100K@\$10 | Price=\$25

##### ARR & Churn:

Jan ARR = \$50K  $\times$  12 = **\$600K annual run rate**

Feb Net New MRR = +\$8K - \$3K = **+\$5K MRR**

Feb Gross Churn Rate =  $\$3K \div \$50K = \mathbf{6\% \text{ per month}}$

Net Rev Churn =  $(\$3K - \$0) \div \$50K = \mathbf{6\%}$  (no expansion data given)

##### CAC & LTV:

CAC =  $(\text{Sales} + \text{Mktg}) \div \text{New Customers} = (\$120K + \$80K) \div 100 = \mathbf{\$2,000 \text{ per customer}}$

LTV =  $(\text{ARPU} \times \text{GM}\%) \div \text{Churn} = (\$500 \times 70\%) \div 0.03 = \$350 \div 0.03 = \mathbf{\$11,667 \text{ per customer}}$

LTV:CAC =  $\$11,667 \div \$2,000 = \mathbf{5.8x}$  ✓ (above 3 = healthy; above 5 = under-investing slightly)

*Intuition: Each customer generates \$11,667 in lifetime gross profit. It cost \$2,000 to acquire them.*

##### EPS & Treasury Stock Method:

Basic EPS =  $\text{Net Income} \div \text{Basic Shares} = \$1M \div 500K = \mathbf{\$2.00}$



